HYEONJUN AN

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EDUCATION

Ulsan National Institute of Science and Technology (UNIST) *Feb 2022 – Present*
Combined M.S. and Ph.D. in Physics; expected February 2027; advisor: Prof. Joonwoo Jeong

Ulsan National Institute of Science and Technology (UNIST) *Mar 2016 – Feb 2022*
B.S., double major in Physics and Energy Engineering; *summa cum laude*

Hong Kong University of Science and Technology (HKUST) *Summer 2017*
Exchange student

RESEARCH AND WORK EXPERIENCE

Ulsan National Institute of Science and Technology (UNIST) Feb 2022 – Present
Doctoral Researcher, Soft Matter Physics Laboratory

- Use H/D neutron radiography to observe hydrogel swelling and quantify solvent redistribution during PDMS swelling and drying with inverse Abel reconstruction and poromechanical modeling.
- Map water-vapor uptake and concentration gradients in hygroscopic droplets and capillaries; investigate gravity, diffusion and mixing.
- Resolve freezing fronts, polymer redistribution and brine networks using synchrotron radiography, tomography and Raman mapping.
- Measure liquid distributions and wetting fronts as evaporative liquids imbibe into heterogeneous porous media.

Pohang Accelerator Laboratory (PAL) Apr–Dec 2024
Research Placement — 50 days at the PLS-II 6C BMI beamline for synchrotron X-ray imaging.

Korea Research Institute of Standards and Science (KRISS) Jan–Feb 2022
Research Intern, Multiscale Length Metrology Team — Femtosecond laser length measurement.

Ulsan National Institute of Science and Technology (UNIST) Mar–Dec 2021
Research Intern, Soft Matter Physics Laboratory — Fluid dynamics in PDMS microfluidic devices.

AWARDS, GRANTS AND HONORS

Poster Presentation Award Nov 2026
HANARO Symposium 2026.

Outstanding Presentation Award in Poster Session Oct 2026
Korean Physical Society Fall Meeting.

First Prize in Poster Session Oct 2026
10th International Topical Meeting on Neutron Radiography (ITMNR-10).

Outstanding Presentation Award in Oral Session Apr 2025; Apr 2026
Korean Physical Society Spring Meetings.

Excellence Award, DRB Space Challenge Aug 2026
Drop-tower microgravity experiment.

Misan Mountaineering Scholarship, UNIST (Fall 2026).

UNIST graduate and teaching assistant scholarships (2021–2026).

National Science and Engineering Undergraduate Scholarship, Korea Student Aid Foundation (2016–2018; 2021).

Principal Investigator, KONICOF 18 Aug 2026 – 17 Feb 2027
2026 Joint Research Program for Ph.D. Students Utilizing Overseas Research Facilities.

Quantitative neutron imaging of mass transport in soft matter with complementary X-ray imaging.

Funder: Korea Nuclear International Cooperation Foundation (KONICOF).

International partner: Paul Scherrer Institute (PSI), Switzerland.

SELECTED PUBLICATIONS

First Author

1. **Hyeonjun An**, B. Kim, J. K. Im, S. Kim, M. W. Kim, S.-G. Kim, J.-H. Lim, K. Kim, J. Jeong. "Polymer-Regulated Freezing of Water Droplets Revealed by Synchrotron X-ray Imaging and Raman Spectroscopy." *Journal of Colloid and Interface Science* 729, 141826 (2027; published online in 2026). DOI: 10.1016/j.jcis.2026.141826.
2. **Hyeonjun An**, Y. Koo, J. Cheon, P. Trtik, J. Jeong. "Condensation of the Vapor in Sessile Water Droplets and Open-Ended Water-Filled Capillaries Revealed by Neutron Radiography." *Springer Proceedings in Physics* 348, 235–243 (2026). DOI: 10.1007/978-3-032-15003-5_27.

Co-Authored

3. H. Yang, G. Ji, M. Choi, S. Park, **Hyeonjun An**, et al. "Suppressed terahertz dynamics of water confined in nanometer gaps." *Science Advances* 10, eadm7315 (2024). DOI: 10.1126/sciadv.adm7315.
4. J. K. Im, **Hyeonjun An**, S.-G. Kim, J.-H. Lim, J. Jeong. "Direct X-ray visualization of bubble-mediated singularity formation in freezing droplets." *Newton*, accepted (2026). arXiv:2511.20571.

EXPERIMENTAL AND ANALYTICAL EXPERTISE

Neutron radiography: 76 beamtime days at ICON and NEUTRA, Paul Scherrer Institute (PSI), Switzerland, and ENF, Korea Atomic Energy Research Institute (KAERI), Korea.

X-ray imaging: 17 beamtime days at PAL PLS-II 6C BMI, 9D and 7C, plus a separate 50-day placement.

Synchrotron scattering and diffraction (SAXS/WAXD): PAL PLS-II 6D and 9A, Korea.

Analysis and modeling: H/D isotope contrast, inverse Abel reconstruction, tomography and segmentation; Flory–Rehner/Darcy and Maxwell–Stefan models.

TEACHING AND SERVICE

Ulsan National Institute of Science and Technology (UNIST) 2018; 2022–2026
Teaching Assistant — General Physics I and II, General Physics Laboratory I, Mathematical Physics, and Thermal and Statistical Physics.

Republic of Korea Air Force, 8th Fighter Wing 2019–2021
Motor transport operator

SELECTED CONFERENCE PRESENTATIONS

Oral Presentations

Korean Physical Society Spring Meetings Apr 2025; Apr 2026
Oral presentations on "Direct Observation of Water Vapor Uptake in Hygroscopic Liquids by Neutron Radiography" (2025) and "Quantifying Polymer Swelling Kinetics via Neutron Radiography" (2026).

Poster Presentations

HANARO Symposium 2026 Nov 2026
Two posters: "Quantifying Polymer Swelling Kinetics via Neutron Radiography" and "Direct Observation of Water Vapor Uptake in Hygroscopic Liquids by Neutron Radiography."

Korean Physical Society Fall Meeting Oct 2026
Poster: "Connectivity of Brine Networks and Tip Morphology in Freezing Saline Droplets."

10th International Topical Meeting on Neutron Radiography (ITMNR-10) Oct 2026
Two posters: "Quantifying Polymer Swelling Kinetics via Neutron Radiography" and "Direct Observation of Water Vapor Uptake in Hygroscopic Liquids by Neutron Radiography."